

■ Unit I: Navigating a Linux Environment

BTE608A — Programming / Bioinformatics Practical

Learning Objectives

By the end of this unit, you should be able to:

- Open and use an Ubuntu/Linux terminal
- Navigate between directories
- Find your current location
- List files and directories
- Create new files and directories
- Copy, move, rename, and remove files and directories
- Execute basic programs from the terminal

1. The Linux Terminal

The terminal allows us to interact with the operating system using commands.

```
command
```

Example:

```
pwd
```

Linux commands are case-sensitive. **ls** and **LS** are different commands.

2. Navigating the File System

pwd — displays your current working directory.

```
pwd
```

ls — lists files and directories in the current location.

```
ls
ls -l
ls -a
```

cd — moves between directories.

```
cd directory_name
cd ..
cd ~
```

3. Creating Files and Directories

mkdir — creates a directory.

```
mkdir bioinformatics
mkdir bioinformatics/practical
```

touch — creates an empty file.

```
touch notes.txt
ls
```

4. Copying Files and Directories

cp — copies files or directories.

```
cp notes.txt notes_backup.txt
cp notes.txt bioinformatics/
cp -r bioinformatics backup
```

5. Moving and Renaming

mv — moves or renames files and directories.

```
mv notes.txt linux_notes.txt
mv linux_notes.txt bioinformatics/
```

6. Removing Files and Directories

rm — removes files. Use **rm -r** to remove a directory and its contents.

```
rm notes.txt
rm -r old_folder
```

■ **Be careful with rm.** Deleted files are generally not moved to a recycle bin.

7. Mini Practical

Create the following directory structure:

```
BTE608A_Unit1/
■■■ notes/
■■■ data/
■■■ backup/
```

Then create **notes/linux_commands.txt** and **data/sample.txt**. Copy **sample.txt** into **backup/**, and rename it to **sample_data.txt**.

8. Practice Questions

Exercise 1 — Find your current directory.

```
pwd
```

Exercise 2 — List all files, including hidden files.

```
ls -a
```

Exercise 3 — Create a directory called unit1.

```
mkdir unit1
```

Exercise 4 — Inside unit1, create data, scripts, and results.

```
mkdir -p unit1/data unit1/scripts unit1/results
```

Exercise 5 — Create an empty file called student_notes.txt.

```
touch student_notes.txt
```

Exercise 6 — Copy student_notes.txt into the data directory.

```
cp student_notes.txt unit1/data/
```

Exercise 7 — Rename student_notes.txt to linux_notes.txt.

```
mv student_notes.txt linux_notes.txt
```

Exercise 8 — Move linux_notes.txt into the notes directory.

```
mkdir -p notes
mv linux_notes.txt notes/
```

9. Final Check ■

Before completing Unit I, you should be comfortable with:

Command	Purpose
pwd	Show current directory
ls	List files and directories
cd	Change directory
mkdir	Create a directory
touch	Create an empty file
cp	Copy
mv	Move or rename
rm	Remove

These commands form the basic foundation for working with Linux in bioinformatics.

What Comes Next?

In Unit II, we will learn how to view and manipulate the contents of files using commands such as **cat**, **less**, **more**, **head**, **tail**, **grep**, **cut**, **paste**, **sed**, and **awk**. We will also learn about piping and writing files using Bash.